

SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID - SIH26241
- Problem Statement Title -
- AI-Enabled Career Counselling and Family Decision-Support Platform for Vocational Education
- Theme - Smart Education
- PS Category - Software
- Team ID -
- Team Name - SmritiCare



SmritiCare

VocGuide

A bilingual, voice-enabled career counselling & family decision-support ecosystem purpose-built for Indian vocational skilling.



The Problem

- Parents are the real decision-makers (80%+)
- Strong cultural bias towards conventional degrees
- Unaddressed anxieties about safety, pay and career progression.
- Existing guidance tools are student-centric, ignoring the veto power.

The Solution

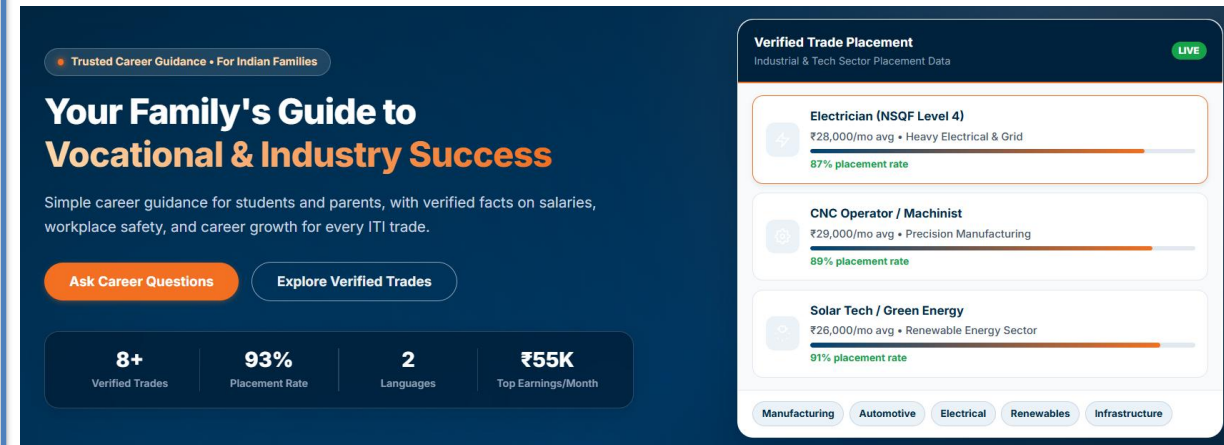
- Joint Conversation Guide
- Uses empirical benchmarks (NCVT, DGT) for real salary & Placement Figures
- Reassures families about safety ratings and PPE standards
- Human Counsellors and real-time dashboard to track resistance.

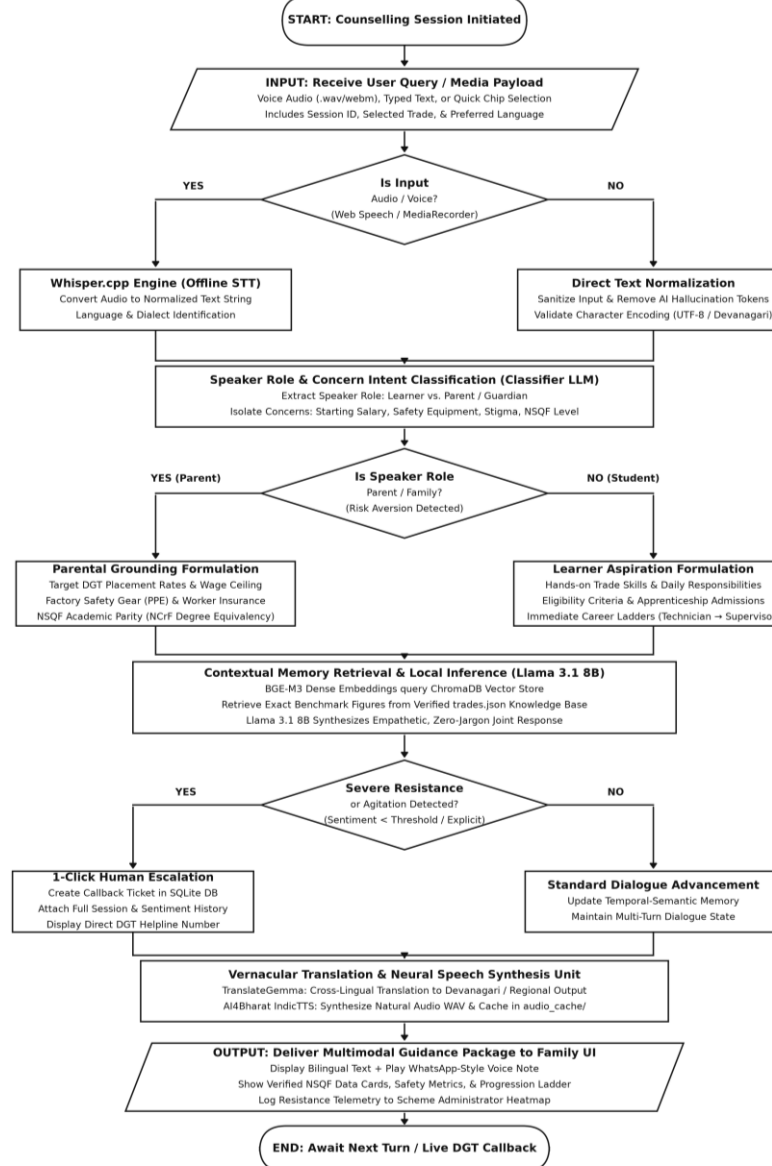
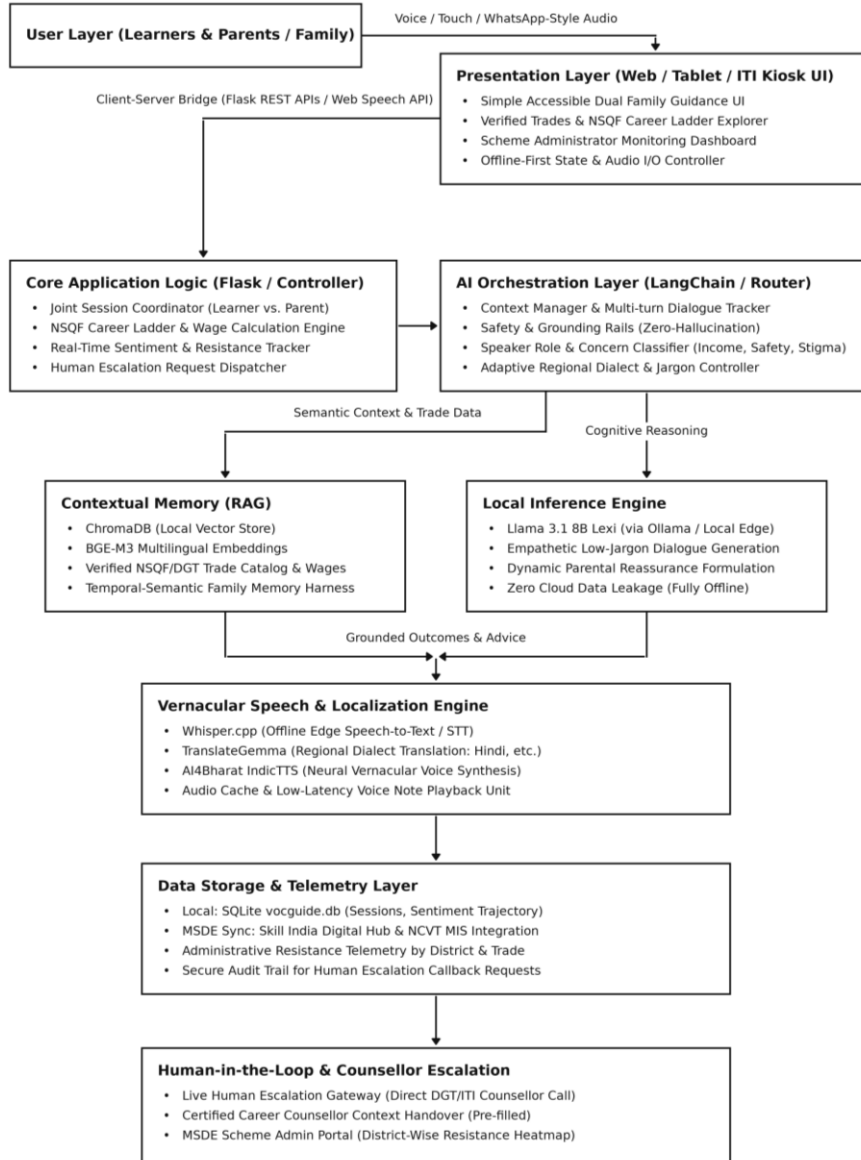


Our Innovation



Dimension	Conventional Career Tools	Our Innovation
Target Audience	Learner-only (ignores family influence).	Joint Family Unit
Data Integrity	Generic AI answers, static articles, or hallucinated claims.	Zero-Hallucination RAG
Accessibility	Text-heavy, English-first, complex dashboards.	Voice-First & Vernacular
Resolution Pathway	Bot terminates when user is unsatisfied.	Live Human Escalation
Government Utility	Black box with zero policy feedback.	Policy Resistance Telemetry





Technology Stack

Frontend

- Vanilla HTML5, CSS3 & JavaScript

Backend

- Python 3.14 + Flask

AI Layer

- LangChain
- Meta Llama 3.1 8B
- TranslateGemma, BGE-M3

Embeddings

Speech & Audio Layer

- Whisper.cpp, AI4Bharat IndicTTS

Local Storage & Database

- SQLite3, ChromaDB

Technical Feasibility

1. **Working End-to-End Prototype.**
2. **Edge & Consumer-Hardware Compatible:** Powered by 4-bit/8-bit quantized models (Meta Llama 3.1 8B, Whisper.cpp, and BGE-M3), the entire **AI stack can run locally on modest consumer hardware.**
3. **Offline-Resilient RAG Architecture:** Retrieval-Augmented Generation (ChromaDB + SQLite) and local MD5 audio caching allow the platform to **serve repeated parent queries with instant <100ms response** times, even in areas with intermittent rural internet connectivity.

Operational & Ecosystem Feasibility

1. **Seamless MSDE / NCVT Integration:** Designed with standard RESTful APIs that easily plug into existing government skilling infrastructure.
2. **Plug-and-Play Institutional Rollout:** Requires zero complex client installation, accessible via a lightweight web browser on smartphones or through kiosks at 15,000+ Industrial Training Institutes (ITIs) and rural Common Service Centres (CSCs).
3. **Low Maintenance Overhead:** Data is decoupled into verified schemas (trades.json / SQLite), allowing non-technical scheme administrators to update wage benchmarks, scheme subsidies, or new trades without touching model code.

Economic & Financial Viability

1. **Zero Recurring API Licensing Fees:** Built entirely on an open-source AI stack eliminating per-token cloud costs.
2. **Ultra-Low Cost per Session:** Estimated compute cost is under ₹0.08 per completed counselling session, making population-scale deployment feasible on government budgets.
3. **High Return on Public Investment (ROI):** In India, a single mid-course ITI dropout costs the public exchequer over ₹45,000 in wasted subsidies. **Preventing just 2, 3 dropouts** per ITI recovers the platform's deployment cost for an entire district.

Impact on Target Audience

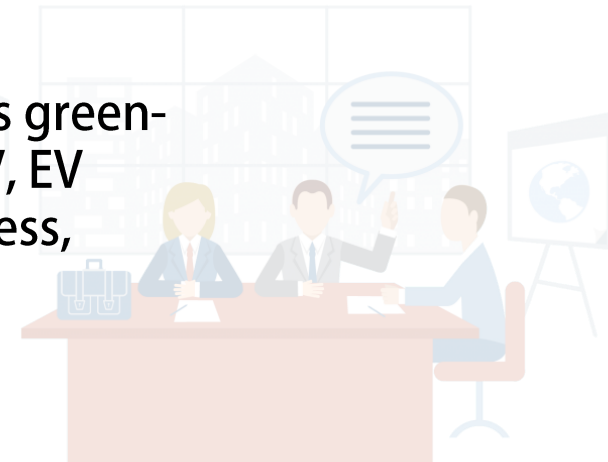
For Youth: Eliminates parental veto; provides clear career ladders (e.g. ₹28k starting -> ₹55k supervisor).

For Parents: Replaces fear of "dead-ends" with 82 - 93% verified placement rates and factory PPE safety proofs.

For MSDE / Admins: Real-time Resistance Heatmap pinpoints exact districts and reasons where parents hesitate.

Solution Benefits

- Social: Restores *Dignity of Labor*, boosts female enrollment in safe, certified trades (Healthcare 93% placement).
- Economic: 35 - 40% drop in mid-course dropouts, saving the public exchequer ₹45,000+ per student in wasted subsidies.
- Environmental: Promotes green-collar pathways (Solar PV, EV Battery) via 100% paperless, remote counselling.



A. ITI outcomes and perception

- [1] Directorate General of Training (DGT), MSDE, "Tracer Study of ITI Graduates: Final Report," Mott MacDonald, New Delhi, Jan. 2018. <https://dgt.gov.in/sites/default/files/pdf/Tracer%20Study%20of%20ITI%20Graduates-Final%20report.pdf>
- [2] DGT, MSDE, "Tracer Study to Assess Employment Outcomes of ITI Pass Outs (STRIVE Project)," Deloitte, New Delhi, May 2024. <https://dgt.gov.in/sites/default/files/National%20Tracer%20Study%20-%20STRIVE.pdf>
- [3] NITI Aayog, "Transforming Industrial Training Institutes," New Delhi, Jan. 2023. https://www.niti.gov.in/sites/default/files/2023-02/ITI_Report_02022023_0.pdf
- [4] U. Ajithkumar and M. Pilz, "Attractiveness of Industrial Training Institutes (ITI) in India: A study on ITI students and their parents," *Education + Training*, vol. 61, no. 2, pp. 153, 168, 2019. <https://doi.org/10.1108/ET-04-2018-0102>

B. Policy

- [5] Ministry of Education, Govt. of India, *National Education Policy 2020*, New Delhi, 2020. https://static.pib.gov.in/WriteReadData/userfiles/NEP_Final_English_0.pdf
- [6] NCVET, MSDE, "Notification of National Credit Framework (NCrF) in respect of Vocational Education," F. No. 26001/01/2019-20/NCVET, 2023. <https://ncvet.gov.in/wp-content/uploads/2023/05/National-Credit-Framework-NCVET-notification.pdf>. Report of the High-Level Committee: https://ncvet.gov.in/wp-content/uploads/2023/05/Report-on-National_Credit_Framework.pdf

C. Speech, language and retrieval

- [7] G. Gerganov and contributors, "whisper.cpp," GitHub. <https://github.com/ggml-org/whisper.cpp>
- [8] AI4Bharat, "*IndicTTS: Text-to-Speech Synthesis in Indian Languages with FastPitch and HiFi-GAN Architectures*," Indian Institute of Technology (IIT) Madras, 2023. [Online]. Available: <https://ai4bharat.org>

D. Stakeholders

- [9] Ministry of Skill Development and Entrepreneurship (MSDE) (<https://www.msde.gov.in>), National Council for Vocational Education and Training (NCVET) (<https://ncvet.gov.in>), NITI Aayog (<https://www.niti.gov.in>), Directorate General of Training (DGT) (<https://dgt.gov.in>), National Council for Vocational Training (NCVT) (<https://ncvtmis.gov.in>).